226184: sandstone, Nicolay 1

(Nambeet Formation, Canning Basin)

Location and sampling

Joanna Spring (SF 51-3), McTavish (3556)
MGA Zone 51, 526953E 7825419N

Sampled on 18 February 2019

This sample was collected from the 3424.7 – 3425.7 m depth interval of drillcore Nicolay 1, from a petroleum well drilled in 2012 by New Standard Onshore Pty Ltd (New Standard Onshore, 2013). The drillhole is located on a north–south cleared line, about 52 km south of Battlement Rocks, 30 km southeast of Turkey Place Hill, and 18.8 km west-northwest of McTavish Claypan.

Geological context

The unit sampled is the Early Ordovician (485–470 Ma; Cohen et al. 2020) Nambeet Formation of the Canning Basin (Haines, 2004). The unit is about 337 m thick in Nicolay 1 and consists mainly of shale with interbedded limestone and sandstone (New Standard Onshore, 2013). The Nambeet Formation is the lowermost Canning Basin unit in Nicolay 1 and is conformably overlain by the Willara Formation (Haines, 2004; Zhan, 2019). Three additional samples of Nambeet Formation sandstone from Nicolay 1 yielded maximum depositional ages of 589–533 Ma (GSWA 226183, Wingate et al., 2021a; GSWA 226185, Wingate et al., 2021b; GSWA 226186, Wingate et al., 2021c). A Nambeet Formation sandstone sample from Samphire Marsh 1, drilled about 246 km to the west-northwest, yielded detrital zircon ages of 2838–506 Ma, and a conservative maximum depositional age of 519 ± 3 Ma (GSWA 135962, Wingate and Haines, 2009). Another Nambeet Formation sample from Barnicarndy 1, drilled about 185 km to the southwest, yielded detrital zircon ages of 3274–590 Ma, and a conservative maximum depositional age of 1007 ± 18 Ma (GSWA 237998, Wingate et al., 2021d).

Petrographic description

The sample is very fine-grained, fossiliferous quartz sandstone (Fig. 1). Excluding carbonate cement, the sample consists of about 70% quartz, 15–20% carbonate grains and fossil fragments, <10% feldspar, 2% mica, and trace tourmaline, zircon, pyrite and glauconite. Although bedding is not obvious, there is subparallel alignment of elongate fossil fragments, detrital micas, and discontinuous, wavy stylolitic folia. Quartz is mainly subangular, clear, unstrained, and up to 0.15 mm in diameter. Feldspar is angular and subangular, up to 0.1 mm in size, and some grains exhibit multiple twinning. Most carbonate grains are up to 0.3 mm in diameter, many display radial texture of probable ooids, and there are minor rod-like grains interpreted as fossil fragments, including a probable shell fragment 8 mm long. Detrital micas include laths of muscovite and rare chlorite up to 0.2 mm long, and green tourmaline is subrounded and up to 0.1 mm in size. Rounded green probable glauconite grains are up to 0.1 mm diameter. Most of the sample contains carbonate cement although there are irregular patches in which the carbonate cement is either iron-stained or contains clay minerals. Pyrite forms disseminated irregular patches up to 0.1 mm diameter, which exhibit framboidal textures, and is also concentrated along stylolite surfaces. The sandstone is well sorted but contains angular grains, and is classified as texturally submature to mature. The presence of fossils, ooids, and glauconite indicates a marine environment. The sandstone is not deformed or metamorphosed.

Zircon morphology

Zircons isolated from this sample are mostly colourless, anhedral to euhedral, and variably rounded. The crystals are up to 350 µm long, and equant to elongate, with aspect ratios up to 4:1. Many crystals have pitted outer surfaces and, in cathodoluminescence (CL) images, concentric zoning truncated at grain boundaries, features consistent with sedimentary transport. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 11 June 2020, using LA-ICP-MS instruments at the GeoHistory Facility in the John de Laeter Centre, Curtin University. Separated zircons mounted in epoxy disks were ablated using a Resonetics RESOlution M-50A-LR sampling system, incorporating a 193 nm Compex 102 excimer laser, and ion signal intensities were measured using an Agilent 8900s quadrupole ICP-MS. Samples were ablated for 22 s at a 5 Hz repetition rate, using a 24 µm beam diameter and laser energy of 2.0 J/cm^2. Zircon standard 91500 was used as the primary reference material, and GJ1, OGC1 and Plešovice were included as secondary standards. Additional analytical details are provided in Fielding et al. (2021). All relevant uncertainties have been propagated into the ratios and dates reported in Table 1. Ages from analyses for which $^{238}U/^{206}Pb$ ratios indicate dates <1000 Ma are based on 207-corrected $^{238}U/^{206}Pb^*$ ratios; those >1000 Ma are based on uncorrected $^{207}Pb/^{206}Pb$ ratios.

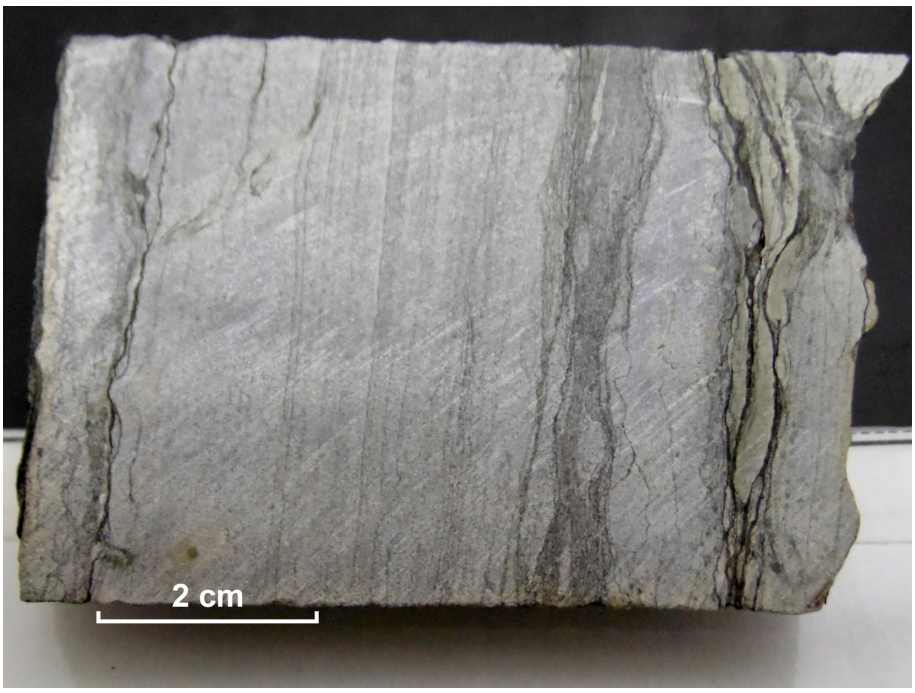

Figure 1. Drillcore image of sample 226184: sandstone, Nicolay 1

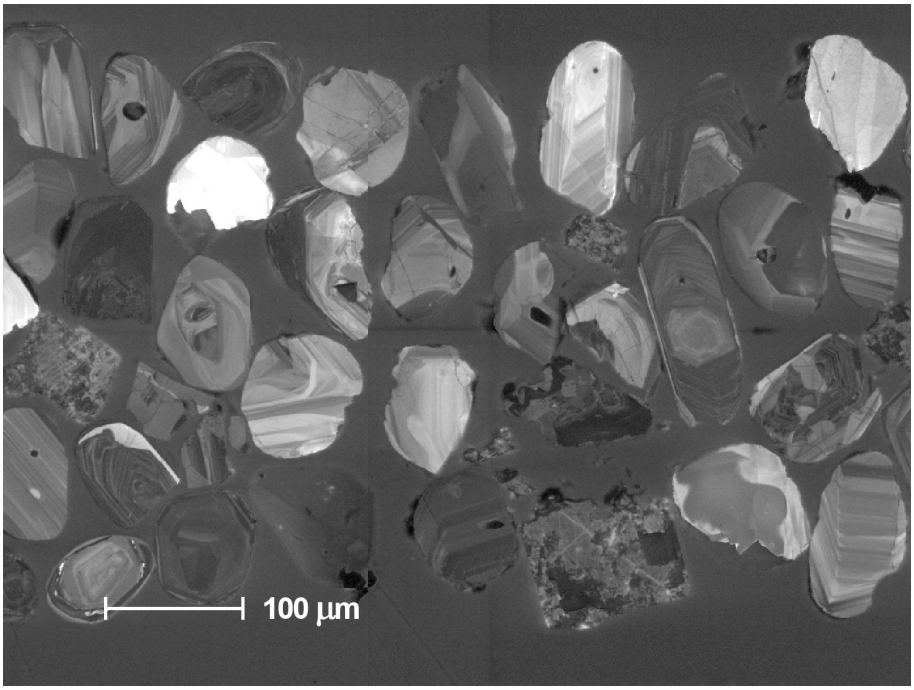

Figure 2. Cathodoluminescence image of representative zircons from sample 226184: sandstone, Nicolay 1

Table 1. LA-ICP-MS analytical results for zircons from sample 226184: sandstone, Nicolay 1

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		Disc. (%)
Y	64	64	71	46	0.64	13.986	0.137	0.05830	0.00235	445	4	541	88	17.7
S	65	65	440	357	0.81	13.554	0.074	0.06150	0.00110	459	2	657	38	30.1
S	82	82	200	91	0.45	13.495	0.100	0.05680	0.00145	461	3	484	56	4.7
S	60	60	137	134	0.98	12.821	0.099	0.05530	0.00140	484	4	424	56	-14.1
S	66	66	209	117	0.56	11.933	0.078	0.05970	0.00120	519	3	593	44	12.5
S	5	5	411	324	0.79	11.794	0.067	0.06070	0.00100	525	3	629	35	16.5
S	21	21	207	81	0.39	11.405	0.065	0.05840	0.00140	542	3	545	52	0.6
S	70	70	147	115	0.78	11.161	0.093	0.05970	0.00160	553	4	593	58	6.7
S	84	84	325	112	0.35	11.044	0.067	0.05900	0.00115	559	3	567	42	1.5
S	20	20	92	191	2.07	10.953	0.090	0.06040	0.00190	563	4	618	68	8.9
S	48	48	233	215	0.92	10.941	0.078	0.06060	0.00120	564	4	625	43	9.8
S	1	1	38	118	3.06	10.893	0.131	0.05890	0.00245	566	6	563	91	-0.5
S	44	44	650	24	0.04	10.870	0.057	0.06040	0.00090	567	3	618	32	8.2
S	53	53	83	82	0.98	10.672	0.091	0.05840	0.00160	577	5	545	60	-6.0
S	2	2	442	92	0.21	10.371	0.053	0.06070	0.00095	593	3	629	34	5.6
S	79	79	271	115	0.43	10.213	0.057	0.06120	0.00110	602	3	646	39	6.8
S	94	94	141	97	0.69	10.070	0.081	0.06050	0.00140	610	5	622	50	1.8
S	71	71	985	243	0.25	10.010	0.049	0.06135	0.00080	614	3	652	28	5.8
S	16	16	630	116	0.18	9.294	0.056	0.06370	0.00090	659	4	732	30	10.0
S	26	26	873	595	0.68	7.123	0.036	0.07200	0.00095	847	4	986	27	14.1
S	28	28	85	17	0.20	6.863	0.061	0.07280	0.00180	877	7	1008	50	13.0
S	8	8	532	387	0.73	6.515	0.047	0.07250	0.00100	921	6	1000	28	7.9
S	7	7	704	577	0.82	6.309	0.034	0.07730	0.00100	948	5	1129	26	16.0
S	3	3	285	254	0.89	6.188	0.034	0.07520	0.00120	966	5	1074	32	10.1
S	10	10	324	225	0.69	6.031	0.033	0.07600	0.00105	989	5	1095	28	9.7
S	69	69	204	44	0.22	5.945	0.032	0.07430	0.00115	1002	5	1050	31	4.5
S	11	11	176	216	1.23	5.640	0.035	0.07460	0.00130	1052	6	1058	35	0.5
S	37	37	375	113	0.30	5.675	0.031	0.07550	0.00110	1046	5	1082	29	3.3
S	75	75	167	70	0.42	5.656	0.034	0.07620	0.00130	1049	6	1100	34	4.6
S	89	89	235	109	0.46	5.432	0.032	0.07660	0.00115	1089	6	1111	30	1.9
S	59	59	57	68	1.20	5.441	0.043	0.07690	0.00195	1088	8	1119	51	2.8
S	55	55	51	107	2.08	5.025	0.039	0.07810	0.00210	1170	8	1149	53	-1.8
S	6	6	176	132	0.75	4.931	0.030	0.08060	0.00125	1190	7	1212	31	1.8

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	Disc. (%)
S	52	52	213	228	1.07	4.904	0.028	0.08080	0.00130	1196	6	1217	32	1.7
S	72	72	110	73	0.66	4.921	0.031	0.08180	0.00155	1192	7	1241	37	3.9
S	23	23	186	156	0.84	4.706	0.027	0.08190	0.00130	1242	6	1243	31	0.1
S	63	63	83	94	1.13	4.585	0.035	0.08240	0.00155	1272	9	1255	37	-1.3
S	56	56	968	54	0.06	4.852	0.025	0.08249	0.00100	1208	6	1257	24	3.9
S	43	43	197	130	0.66	4.369	0.026	0.08440	0.00120	1329	7	1302	28	-2.1
S	57	57	323	219	0.68	4.301	0.024	0.08540	0.00115	1348	7	1325	26	-1.7
S	73	73	121	56	0.46	4.209	0.025	0.08800	0.00145	1374	7	1382	32	0.6
S	76	76	294	124	0.42	4.062	0.022	0.09080	0.00125	1419	7	1442	26	1.6
S	93	93	233	143	0.61	3.817	0.022	0.09250	0.00130	1500	8	1478	27	-1.5
S	17	17	669	477	0.71	3.998	0.020	0.09310	0.00110	1439	6	1490	22	3.4
S	61	61	201	193	0.96	3.839	0.020	0.09490	0.00135	1492	7	1526	27	2.2
S	47	47	573	34	0.06	3.679	0.019	0.09850	0.00120	1550	7	1596	23	2.9
S	51	51	516	70	0.14	3.564	0.018	0.09870	0.00120	1594	7	1600	23	0.3
S	30	30	539	227	0.42	3.715	0.018	0.09910	0.00120	1537	7	1607	23	4.4
S	62	62	213	106	0.50	3.421	0.019	0.10040	0.00125	1653	8	1632	23	-1.3
S	31	31	223	118	0.53	3.232	0.018	0.10940	0.00145	1738	9	1789	24	2.9
S	40	40	223	144	0.64	2.911	0.014	0.11840	0.00150	1903	8	1932	23	1.5
S	42	42	405	35	0.09	2.618	0.013	0.13000	0.00160	2085	9	2098	22	0.6
S	83	83	124	82	0.66	2.313	0.012	0.14300	0.00185	2317	10	2264	22	-2.3
S	74	74	908	90	0.10	2.386	0.011	0.14400	0.00160	2256	9	2276	19	0.9
S	85	85	416	175	0.42	2.107	0.010	0.16350	0.00190	2504	10	2492	20	-0.5
S	36	36	243	177	0.73	2.035	0.010	0.17760	0.00210	2577	10	2631	20	2.0
S	27	27	210	79	0.38	2.070	0.010	0.17920	0.00220	2541	10	2645	20	4.0
S	38	38	76	48	0.63	1.956	0.012	0.18730	0.00245	2662	13	2719	22	2.1
S	81	81	23	21	0.92	1.904	0.018	0.19660	0.00305	2721	21	2798	25	2.8
S	54	54	117	107	0.91	1.771	0.012	0.21330	0.00255	2886	16	2931	19	1.5
S	33	33	139	99	0.71	1.632	0.008	0.25610	0.00300	3081	12	3223	18	4.4
S	46	46	26	23	0.91	1.593	0.013	0.26080	0.00400	3140	20	3252	24	3.4
D	32	32	115	38	0.33	5.288	0.057	0.07220	0.00145	1116	11	992	41	-12.6
D	87	87	142	59	0.42	5.562	0.037	0.07230	0.00140	1066	7	994	39	-7.2
D	77	77	23	13	0.56	5.914	0.059	0.07570	0.00325	1007	9	1087	86	7.4
D	15	15	695	107	0.15	5.807	0.037	0.07580	0.00095	1024	6	1090	25	6.0

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		Disc. (%)
D	9	9	361	376	1.04	5.714	0.029	0.07790	0.00105	1040	5	1144	27	9.2
D	12	12	515	88	0.17	5.402	0.025	0.07900	0.00115	1095	5	1172	29	6.6
D	92	92	389	222	0.57	5.365	0.027	0.07950	0.00110	1102	5	1185	27	7.0
D	50	50	233	45	0.19	5.513	0.029	0.08030	0.00130	1075	5	1204	32	10.8
D	18	18	528	204	0.39	5.110	0.027	0.08160	0.00110	1152	6	1236	26	6.8
D	58	58	406	188	0.46	5.000	0.069	0.09080	0.00120	1175	15	1442	25	18.5
D	19	19	126	140	1.11	4.117	0.028	0.09300	0.00145	1402	9	1488	30	5.8
D	80	80	210	105	0.50	4.769	0.052	0.09400	0.00135	1227	12	1508	27	18.6
D	67	67	150	41	0.28	4.525	0.025	0.09640	0.00155	1287	6	1556	30	17.3
D	86	86	395	42	0.11	3.789	0.019	0.09830	0.00120	1510	7	1592	23	5.2
D	68	68	512	225	0.44	3.940	0.023	0.09890	0.00125	1458	7	1603	24	9.1
D	35	35	476	170	0.36	3.794	0.021	0.09990	0.00120	1508	7	1622	22	7.0
D	14	14	325	185	0.57	3.867	0.024	0.10210	0.00130	1483	8	1663	24	10.8
D	45	45	74	44	0.60	4.296	0.040	0.10500	0.00330	1349	11	1714	58	21.3
D	39	39	301	150	0.50	3.568	0.019	0.10870	0.00145	1593	8	1778	24	10.4
D	22	22	173	158	0.91	3.399	0.020	0.11110	0.00150	1662	8	1817	25	8.5
D	96	96	1421	60	0.04	3.226	0.016	0.11890	0.00135	1741	7	1940	20	10.3
D	13	13	328	74	0.23	3.284	0.018	0.11930	0.00150	1714	8	1946	22	11.9
D	4	4	1095	104	0.09	4.429	0.025	0.12220	0.00145	1312	7	1989	21	34.0
D	95	95	215	108	0.50	3.050	0.017	0.12450	0.00155	1828	9	2022	22	9.6
D	29	29	508	262	0.52	3.488	0.016	0.13600	0.00160	1625	7	2177	20	25.4
D	34	34	289	58	0.20	2.851	0.017	0.15170	0.00185	1938	10	2365	21	18.1
D	90	90	183	65	0.35	2.345	0.013	0.15990	0.00200	2289	11	2455	21	6.7
D	41	41	121	28	0.23	2.251	0.014	0.16480	0.00220	2369	13	2506	22	5.4
D	24	24	396	449	1.13	2.375	0.013	0.17020	0.00200	2265	11	2560	20	11.5
D	91	91	266	179	0.67	2.571	0.046	0.17110	0.00215	2118	32	2568	21	17.5
D	49	49	826	776	0.94	2.234	0.011	0.17900	0.00200	2385	10	2644	19	9.8
D	97	97	669	702	1.05	4.042	0.020	0.18160	0.00205	1425	6	2668	19	46.6
D	88	88	725	152	0.21	2.143	0.011	0.18810	0.00215	2469	11	2726	19	9.4
D	25	25	104	44	0.42	2.030	0.011	0.19380	0.00250	2581	12	2775	21	7.0
D	78	78	1591	146	0.09	2.128	0.010	0.19790	0.00220	2484	9	2809	18	11.6

Results

Ninety-seven analyses were obtained from 97 zircons. Results are listed in Table 1, and shown in a concordia diagrams (Fig. 3) and probability density plots (Figs 4, 5).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Thirty-five analyses >1000 Ma are >5% discordant. The dates obtained from these 35 analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 62 analyses can be divided into two groups, based on their $^{207}Pb/^{206}Pb$ and $^{238}U/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a 207-corrected $^{238}U/^{206}Pb^*$ date of 444 ± 4 Ma (1σ).

Group S comprises 61 analyses (Table 1), which yield $^{207}Pb/^{206}Pb$ or 207-corrected $^{238}U/^{206}Pb^*$ dates of 3252–456 Ma.

It is possible that all of the analyses are of unmodified detrital zircons, in which case the date of 444 ± 4 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age for the siltstone. A more conservative estimate of the maximum depositional age can be based on the weighted mean 207-corrected $^{238}U/^{206}Pb^*$ date of 562 ± 5 Ma (MSWD = 1.6) for the youngest coherent group of six analyses in Group S.

The data for combined Groups Y and S indicate significant age components at c. 1606, 1487, 1243, 1088, and 563 Ma, based on contributions from approximately 4, 4, 7, 7, and 5 analyses, respectively (Fig. 5). These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or as the ages of detrital components within sediments that have been reworked into this rock.

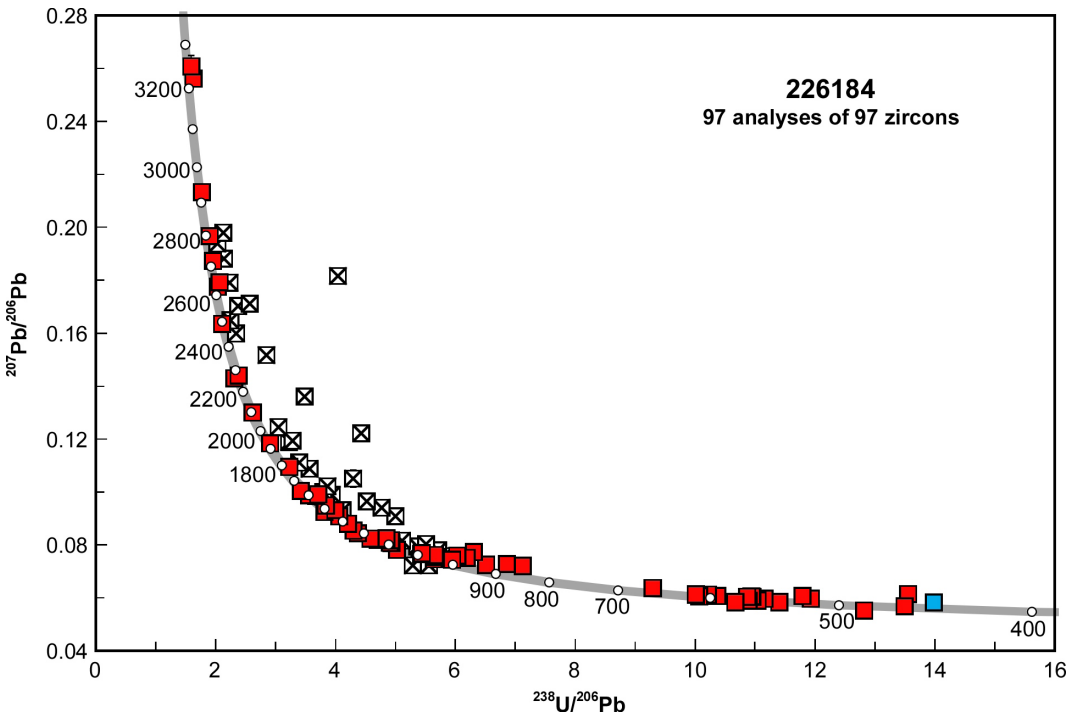

Figure 3. U–Pb analytical data, not corrected for common Pb, for zircons from sample 226184: sandstone, Nicolay 1. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed squares indicate Group D (date >1000 Ma and discordance >5%)

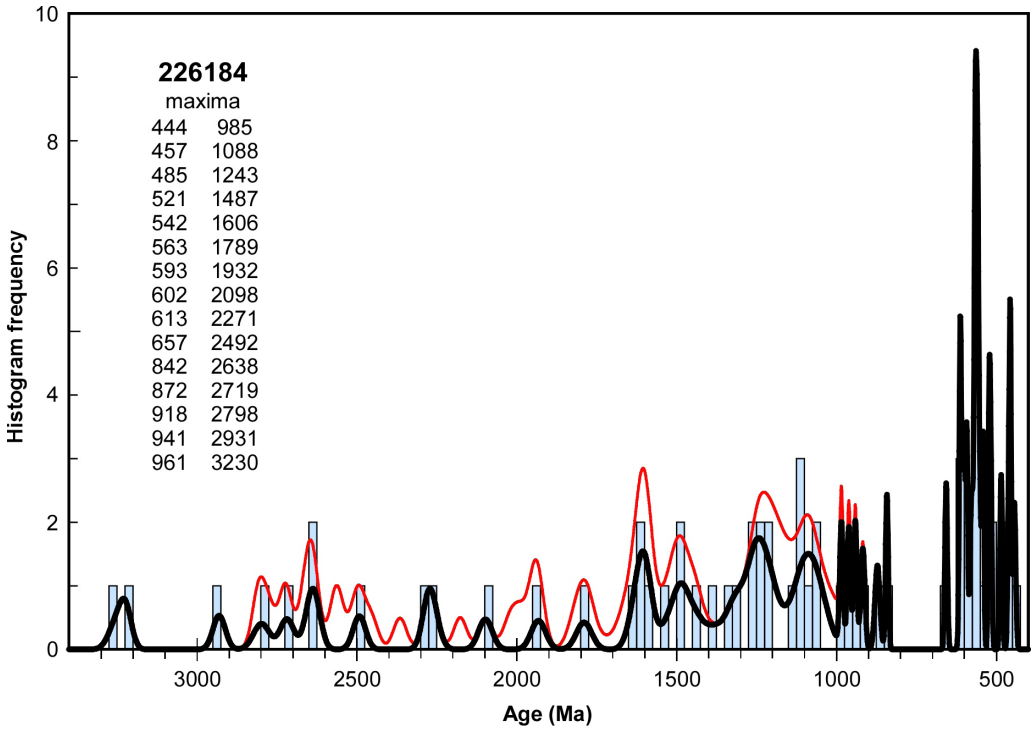

Figure 4. Probability density diagram and histogram for sample 226184: sandstone, Nicolay 1. Thick curve, maxima values, and frequency histogram (bin width 25 Ma) include only data in Groups Y and S (62 analyses of 62 zircons). Thin curve includes all data (97 analyses of 97 zircons)

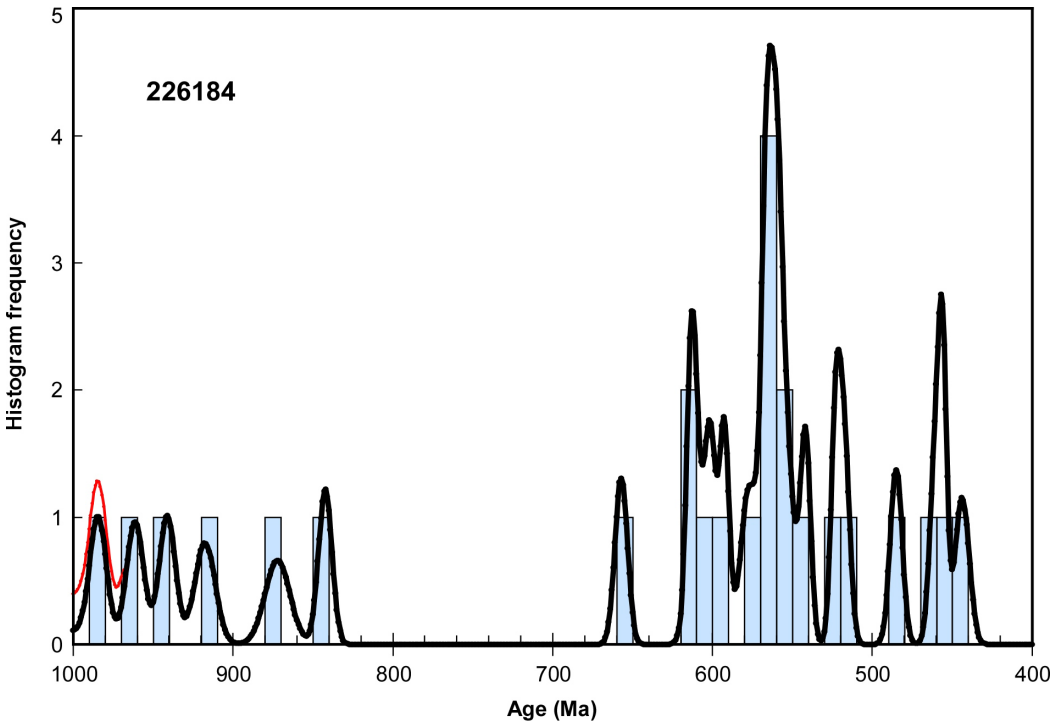

Figure 5. Expanded view of probability density diagram and histogram for sample 226184: sandstone, Nicolay 1. Bin width is 10 Ma

References

Cohen, KM, Finney, SC, Gibbard, PL and Fan, J-X 2020, The ICS International Chronostratigraphic Chart (v. 2020/03 update): Episodes, v. 36 (2013), p. 199–204, <https://stratigraphy.org/chart>.

Haines, PW 2004, Depositional facies and regional correlations of the Ordovician Goldwyer and Nita Formations, Canning Basin, Western Australia, with implications for petroleum exploration: Geological Survey of Western Australia, Record 2004/7, 45p.

New Standard Onshore 2013, Nicolay-1, Well Completion Report (Basic Data): Geological Survey of Western Australia, 859p. (unpublished).

Wingate, MTD and Haines, PW 2009, 135962: sandstone, Samphire Marsh 1; Geochronology Record 777: Geological Survey of Western Australia, 5p.

Wingate, MTD, Lu, Y and Dent, LM 2021a, 226183: sandstone, Nicolay 1; Geochronology Record 1760: Geological Survey of Western Australia, 7p.

Wingate, MTD, Lu, Y and Dent, LM 2021b, 226185: sandstone, Nicolay 1; Geochronology Record 1762: Geological Survey of Western Australia, 8p.

Wingate, MTD, Lu, Y and Dent, LM 2021c, 226186: sandstone, Nicolay 1; Geochronology Record 1763: Geological Survey of Western Australia, 5p.

Wingate, MTD, Lu, Y, Fielding, IOH, Normore, LS and Haines, PW 2021d, 237998: quartz sandstone, Barnicarndy 1; Geochronology Record 1743: Geological Survey of Western Australia, 8p.

Zhan, Y 2019, A seismic interpretation of the Broome Platform, Willara Sub-basin and Munro Arch of the Canning Basin, Western Australia: Geological Survey of Western Australia, Report 193, 43p.

Recommended reference for this publication

Wingate, MTD, Lu, Y and Dent, LM 2021, 226184: sandstone, Nicolay 1; Geochronology Record 1761: Geological Survey of Western Australia, 8p.

Data obtained: 11 June 2020

Data released: 14 May 2021